Alex Stott

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Experience

2026 - : Postdoc, Euskal Herriko Unibertsitatea, (*Bilbao, Spain*)
Comparative planetology through atmospheric dynamics. Looking at circulation, turbulence and surface-atmospheric coupling.

2025 - 2025: Post Doctorant, ISAE-Supaero, Université de Toulouse, (*Toulouse, France*)
Created a machine learning detection and analysis framework for seismic and atmospheric events in stratospheric balloon flight pressure data on Earth in preparation for Venus.

2023 - 2025: CNES Research Fellow, ISAE-Supaero, Université de Toulouse, (*Toulouse, France*)
Title: "A machine learning framework for geophysical and atmospheric monitoring in planetary science missions."
Developed machine learning tools for Martian environmental seismology, leading to placing constraints on Mars' winds inter-annual variation at InSight. Created a re-classification of Martian seismic events to determine source and crustal variations. Performed the first characterisation of the dissipative regime of turbulence on Mars using microphone data.

2020 - 2022: Post Doctorant, ISAE-Supaero, Université de Toulouse, (*Toulouse, France*)
Post on seismology and atmospheric science for the NASA InSight and Perseverance missions. *On InSight:* developed machine learning to detect and analyse marsquakes. *On Perseverance:* characterised drivers of turbulent wind fluctuations.

2018 - 2020: Pre/Postdoctoral Research Associate, Imperial College London, (*London, UK*)
Analysed seismic data and instrument performance for the NASA InSight mission. Included a 3 month stay at NASA Jet Propulsion Laboratory for InSight mission deployment operations.

2013 - 2013: Engineering Intern (6 months) at BAE systems Detica, (*Guildford, UK*)
Developed hardware for conversion of high-speed ethernet signal protocols for use in data centres.

Education

2014 - 2019: PhD in Signal Processing and Planetary Instrumentation, Imperial College London, (*UK*)
Title: "Latent variable regression and applications to planetary seismic instrumentation." Part of the development team for the short period microseismometer for the NASA InSight mission. Quantified instrument performance through testing and developed novel thermal decorrelation technique. Created new algorithms for component analysis with multidimensional algebras.

2010 - 2014: MEng (First class) in Electrical and Electronic Engineering, Imperial College London, (*UK*)
Masters thesis: "Reconstruction of 3D topography with images from the NASA Phoenix lander's Optical Microscope." Specialisms in signal processing, control and power electronics.

Space mission involvement

Mission Leadership:

2022 - : Co-lead of the atmospheric working group for the Mars 2020 Perseverance mission: Responsible for the management of atmospheric operations and coordination of science activities across multiple instrument teams

2024 - : Co-Principal Investigator of Platform Atmospheric Characterisation Instrument Suite for ESA ExoMars mission: Responsible for developing science objectives, the concept of operations and implementing requirements with engineering team. Aided funding acquisition and review board procedures.

Science team membership:

2015-2025: NASA InSight mission: Member of seismic events, impacts, data processing and atmospheric working groups. Led a seismic noise discussion theme.

2020- : NASA Perseverance mission: Member of acoustics and atmospheric working groups. Also member of Diversity, Equity, Inclusion and Accessibility working group.

Mission Operations and development:

NASA Perseverance mission: Acted as Mars 2020 atmospheric science uplink representative in operations shifts (2022-Present). Co-led new automated atmospheric operations methodology to adapt to staffing changes (2024). Designed operations activities to take microphone recordings to observe winds (2022).

NASA InSight mission: Performed role of instrument expert. Responsibilities included providing an analysis of the sensor data downlink for performance and liaising with CNES and JPL teams.

Venus: *VIVA Core team*: Venus orbiter to detect airglow from seismic events and atmospheric dynamics proposed for ESA Fast class (unsuccessful). *Long-lived lander*: Contributing noise modelling for surface seismometer. Team led by NASA Glenn.

Teaching Experience and Presentations

Key Presentations

Invited talks/seminars: on machine learning for seismic events on Mars at IPGP, Paris and on the Perseverance microphone at the Acoustical Society of America
International conference presentations: including EPSC, MAMO, BPSC and AGU.
Science team meetings: presented on research, overview and programmatic topics to InSight and Perserance science teams.
School presentations: to children from a disadvantaged area on Perseverance and its role in studying Mars's habitability.

Supervision

2021-2024 **PhD, ISAE-Supaero** Co-supervision of a PhD student on planetary acoustics and atmospheres (published results).

2021-2025 **"Fin d'étude" internships, ISAE-Supaero** Supervised a 4-month internship on machine learning to detect events from stratospheric balloon flight pressure data, a 5-month internship on machine learning for InSight seismic data and assisted in the supervision of a student for marsquake seismicity interpretation. Each led to published results.

2023 **Second year internship, ISAE-Supaero** 3-month undergraduate internship on machine learning for the classification of moonquakes.

2021-2022 **Masters projects, ISAE-Supaero** Assisted in the supervision of three students (two professional and one post-graduate) masters project on seismometer noise determination for use on small bodies. Provided machine learning expertise for project on asteroid surface properties from experimental impact data.

2017-2020 **Masters projects, Imperial College** Supervision of projects on machine learning for detection of artefacts in InSight seismic data, detection of dust devils in seismic data and using drone camera footage for navigation.

Other teaching activities

2025 **External examiner at University of Oslo** Master's thesis on water vapour retrievals with SuperCam on Mars.

2022&2024 **Co-administered a seismology oral exam, Université de Toulouse.** Required to communicate with the students to demonstrate their knowledge.

2016-2018 **Teaching assistant at Imperial College London** Developed coursework module for Signal processing and machine intelligence course. Administered coursework and problem classes for Advanced Signal Processing and Real-Time Digital Signal Processing courses.

Income and Grants

2026 **ISSI international team** Led proposal for "A comparative planetology study of the atmospheric planetary boundary layer" team at the International Space Science Institute. **Amount:** two week meeting attendance for 15 participants

2025 **Mars 2020 Perseverance Extended Mission** Responsible for the atmospheric science component of the NASA senior review submission for the extended mission. Also contributed to section on preparing for future human missions. **Amount:** Not directly assessable

2024 **ESA ExoMars PACIS Instrument** Co-Principal Investigator for the Platform Atmospheric Characterisation Instrument Suite (PACIS). Meteorology and microphone instrument for the entry, descent and landing stage of the ExoMars mission. **Amount: 600,000 euros**

2022 **CNES (French Space Agency) research fellowship** Competitively funded proposal to create a machine learning framework for geophysical and atmospheric monitoring on planetary missions. **Amount: ∼80,000 euros (2 years)**

Awards

2026 **JPL team award** For exceptional achievement preparing the Mars 2020 Extended Mission proposal and supporting the 2026 Planetary Mission Senior Review process.

2026 **NASA Honor Group Achievement Award** As co-lead of Perseverance Atmospheric Working group for contributions to atmospheric dynamics on Mars.

2025 **JGR: Planets Editor's Highlight** For publication "WindSightNet: the inter-annual variability of Martian winds retrieved from InSight's seismic data with machine learning". (Click here to read). Awarded to only 2% of all AGU journal papers.

2022 **NASA Honor Group Achievement Award** Awarded as part of the Mars 2020 Perseverance atmospheric working group for contributions to mission operations and science.

2017 **IEEE Signal Processing Society Student Travel Grant** For travel to the ICASSP conference in New Orleans, USA.

Professional service

2025 **UK space agency frontiers 2035** Co-authored white papers on Martian weather network, geophysics and Venus cloud balloons.

2024&2025 **Conference convenor** European Geophysical Union Dynamics, tectonics and evolution of rocky planets from formation to the present day. Chair for Data processing session at Mars interior and geophysics after InSight.

2024 **NASA FINESST grant Panellist** Led proposal reviews, communicated with the group and wrote feedback.

2018- **Peer reviewer** Review articles for journals including GRL, Icarus, GJI, BSSA, Signal Processing and PSJ.

Outreach

I have engaged in outreach and media to add impact to several publications. Wrote a "behind the paper" commentary (Click to read) for **The sound of a Martian dust devil (2023)** and gave a live interview for radio New Zealand. Created press materials (audio, text and figures) for the **First sounds from Perseverance's microphone (2021)**, including the Ingenuity rotorcraft. Interviewed for Sky and Telescope magazine and worked with media experts to record an explainer video (Click to watch). Engaged in general outreach including **school talks** and exhibiting on impacts at a **Natural History Museum Late** including participation in "Meet with a scientist".

Relevant voluntary work

Vice Chair, Head of Technical, e.quinox (Oct 2011 - May 2014): Imperial's e.quinox society aims to provide a blueprint for sustainable electricity to rural areas in Africa via renewable energy solutions. Developed a data logger to monitor a solar charging kiosk. Implemented this during participation in a summer 2012 expedition to Rwanda.

Publications

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Lead author or significant role in conceptualisation, development and analysis

1. **A. E. Stott**, L. Margerin, R. F. Garcia, M. Calvet, et al. "A re-classification of Marsquakes based on P-wave coda" *Submitted.* 10.22541/essoar.15008211/v1
2. **A. E. Stott**, S. Gerier, R. F. Garcia, D. Mimoun, T. Heydon and A. Hertzog "A Machine Learning Framework for Event Detection and Analysis in Stratospheric Balloon Flight Data" *Submitted* DOI: 10.22541/essoar.15005338/v1
3. **A. E. Stott**, N. Murdoch, M. Gillier, D. Mimoun, et al. "The dissipation regime of turbulence on Mars with Microphone data from the Perseverance rover" *GRL* (2026) DOI: 10.1029/2025GL117975
4. **A. E. Stott**, R. F. Garcia et al. "WindSightNet: the inter-annual variability of Martian winds retrieved from InSight's seismic data with machine learning" *JGR: Planets* (2025) DOI: 10.1029/2024JE008695
5. **A. E. Stott**, R. F. Garcia, A. Chédozeau, A. Spiga, N. Murdoch, et al. "Machine learning and marsquakes: a tool to predict atmospheric-seismic noise for the NASA InSight mission." *Geophysical Journal International* (2023) DOI: 10.1093/gji/ggac464
6. **A. E. Stott**, N. Murdoch, M. Gillier, D. Banfield, T. Bertrand et al. "Wind and turbulence observations with the Mars microphone on Perseverance." *JGR: Planets* (2023) DOI: 10.1029/2022JE007547.
7. **A. E. Stott**, C. Charalambous, T. J. Warren, W. T. Pike et al. "The Site Tilt and Lander Transfer Function from the Short-Period Seismometer of InSight on Mars " *BSSA* (2021) DOI: 10.1785/0120210058.
8. **A. E. Stott**, C. Charalambous, T. J. Warren, and W. T. Pike. "Full-Band Signal Extraction From Sensors in Extreme Environments: The NASA InSight Microseismometer." *IEEE Sensors Journal* (2018) DOI: 10.1109/JSEN.2018.2871342.
9. **A. E. Stott**, S. Kanna, and D. P. Mandic. "Widely Linear Complex Partial Least Squares for Latent Subspace Regression." *Signal Processing* (2018) DOI: 10.1016/j.sigpro.2018.06.018.
10. **A. E. Stott**, B. Scalzo Dees, I. Kisil, and D. P. Mandic. "A class of multidimensional NIPALS algorithms for quaternion and tensor partial least squares regression." *Signal Processing* (2019) DOI: 10.1016/j.sigpro.2019.03.002.
11. **A. E. Stott**, S. Kanna, D. P. Mandic, and W. T. Pike. "An online NIPALS algorithm for Partial Least Squares." In: Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), *IEEE*, (2017) DOI: 10.1109/ICASSP.2017.7952943.
12. M. Gillier, A. Petculescu, N. Murdoch, **A. E. Stott** et al. "Geographical, seasonal and diurnal variations of acoustic attenuation and sound speed in the near-surface Martian atmosphere" *JGR: Planets* (2024) DOI: 10.1029/2023JE008257.
13. M. Gillier, A. Petculescu, **A. E. Stott**, N. Murdoch et al. "Acoustic propagation in the near-surface Martian atmosphere" *JGR:Planets* (2024) DOI: 10.1029/2024JE008469.
14. N. Murdoch, **A. E. Stott**, D. Mimoun, B. Pinot, A. Chatain et al. "Investigating Diurnal and Seasonal Turbulence Variations of the Martian Atmosphere Using a Spectral Approach." *The Planetary Science Journal* (2023) DOI: 10.3847/PSJ/ad06a9.
15. N. Murdoch, **A. E. Stott**, M. Gillier, R. Hueso, et al. "The sound of a Martian dust devil." *Nature Communications* (2022) DOI: 10.1038/s41467-022-35100-z.
16. C. Charalambous, **A. E. Stott**, W. T. Pike et al. "A comodulation analysis of atmospheric energy injection into the ground motion at InSight, Mars" *JGR: Planets* (2021) DOI: 10.1029/2020JE006538.

Significant role in project development and analysis/interpretation

17. L. Margerin, J. Marti, R. Garcia, **A. E. Stott** et al. "Strongly non-linear elastic behavior of the Martian regolith from two years of seismic noise spectroscopy" *Submitted* (2026)
18. A. Munguira, G. Martinez, A. Sanchez-Lavega, R. Hueso, T. Bertrand, **A. E. Stott** "Surface layer turbulence scaling parameters at Jezero Crater, Mars" *Icarus* (2026): 117332. DOI: 10.1016/j.icarus.2026.117332
19. R. F. Garcia, I. Clemente, M. Drilleau, **A. E. Stott** et al. "Detection of Mars' quake doublets in InSight data" *Journal of Geophysical Research: Planets* (2025) DOI: 10.1029/2024JE008782.
20. I. Daubar, R. F. Garcia, **A. E. Stott**, B. Fernando, G. S. Collins et al. "Seismically detected cratering on Mars: Enhanced recent impact flux ?" *Science Advances* (2024) DOI: 10.1126/sciadv.adk7615.
21. B. Pinot, D. Mimoun, N. Murdoch et al. "The In Situ Evaluation of the SEIS Noise Model." *Space Science Reviews* (2024) DOI: 10.1007/s11214-024-01056-3.
22. D. Mimoun, A. Cadu, N. Murdoch, A. Sournac, Y. Parot, P. Bernardi, P. Pilleri, **A. E. Stott**, et al. "The Mars Microphone onboard SuperCam" *Space science reviews* (2023) DOI: 10.1007/s11214-022-00945-9.
23. B. Fernando, I. J. Daubar, C. Charalambous, P. M. Grindrod, **A. E. Stott**, et al. "A tectonic origin for the largest marsquake observed by InSight." *GRL*, (2023) DOI: 10.1029/2023GL103619.
24. R. D. Lorenz, S. Maurice, B. Chide, D. Mimoun, **A. E. Stott** et al. "The sounds of a helicopter on Mars." *Planetary and Space Science* (2023) DOI: 10.1016/j.pss.2023.105684.
25. S. Maurice, B. Chide, N. Murdoch, R. Lorenz, D. Mimoun, R. Wiens, **A. E. Stott**, X. Jacob, et al. "In situ recording of Mars soundscape." *Nature* (2022) DOI: 10.1038/s41586-022-04679-0.
26. C. Newman, R. Hueso, M. T. Lemmon, et al. "The dynamic atmospheric and aeolian environment of Jezero crater, Mars." *Science Advances* (2022) DOI: 10.1126/sciadv.abn3783.
27. B. Fernando, N. Wojcicka, R. Maguire, S. C. Stähler, **A. E. Stott**, S. Ceylan, et al. "Seismic constraints from a Mars impact experiment using InSight and Perseverance." *Nature Astronomy* (2022) DOI: 10.1038/s41550-021-01502-0.

28. B. Fernando, N. Wojcicka, Z. Han, **A. E. Stott**, S. Ceylan, et al. "Questions to Heaven." *Astronomy & Geophysics* (2021) DOI: 10.1093/astrogeo/atab103.
29. P. Lognonné, W. B. Banerdt, W. T. Pike, D. Giardini, U. Christensen, R. F. Garcia, T. Kawamura et al. "Constraints on the shallow elastic and anelastic structure of Mars from InSight seismic data." *Nature Geoscience* (2020) DOI: 10.1038/s41561-020-0536-y.
30. D. Giardini, P. Lognonné, W. B. Banerdt, W. T. Pike, U. Christensen, S. Ceylan, J. F. Clinton et al. "The seismicity of Mars." *Nature Geoscience* (2020) DOI: 10.1038/s41561-020-0539-8.
31. D. Banfield, A. Spiga, C. Newman, F. Forget, M. Lemmon, R. Lorenz, N. Murdoch et al. "The atmosphere of Mars as observed by InSight." *Nature Geoscience* (2020) DOI: 10.1038/s41561-020-0534-0.
32. Scholz, John-Robert, Rudolf Widmer-Schnidrig, Paul Davis, et al. "Detection, analysis, and removal of glitches from InSight's seismic data from Mars." *Earth and Space Science* (2020) DOI: 10.1029/2020EA001317.
33. M. P. Panning, W. T. Pike, P. Lognonné, W. B. Banerdt, N. Murdoch et al. "On-deck seismology: Lessons from InSight for future planetary seismology." *JGR: Planets* (2020) DOI: 10.1029/2019JE006353.
34. P. Lognonné, W. B. Banerdt, D. Giardini, W. T. Pike, U. Christensen, Phillipe Laudet, S. de Raucourt et al. "SEIS: the seismic experiment for internal structure of InSight." *Space Sci. Rev* (2018) DOI: 10.1007/s11214-018-0574-6.
35. H. Liu, W. T. Pike, C. Charalambous, and **A. E. Stott**. "Passive Method for Reducing Temperature Sensitivity of a Microelectromechanical Seismic Accelerometer for Marsquake Monitoring Below 1 Nano-g." *Physical Review Applied* (2019) DOI: 10.1103/PhysRevApplied.12.064057.
36. M. Xiang, S. Enshaeifar, **A. E. Stott**, C. C. Took, Y. Xia, S. Kanna, and D. P. Mandic. "Simultaneous diagonalisation of the covariance and complementary covariance matrices in quaternion widely linear signal processing." *Signal Processing* (2018) DOI: 10.1016/j.sigpro.2018.02.018.

Contributed particular support, analysis, interpretation and data

35. M. Coimbra, M. de la Torre Juarez, B. McKeon, et al. "Spectral location for the universal scaling regime in Martian atmospheric turbulence" *Nature Comms. Earth and Environment* (2024) DOI: 10.1038/s43247-024-01752-6.
36. C. Robin, N. Murdoch, A. Duchene, et al. "Mechanical properties of rubble pile asteroids: insights from a morphological analysis of surface boulders" *Nature Comms.* (2024) DOI: 10.1038/s41467-024-50147-w.
37. B. Chide, P. Blanc-Benon, T. Bertrand, X. Jacob, et al. "An acoustic investigation of the near-surface turbulence on Mars." *The Journal of the Acoustical Society of America* (2024) DOI: 10.1121/10.0024347.
38. A. Vicente-Retortillo, G. M. Martínez, M. T. Lemmon, R. Hueso, J. R. Johnson, R. Sullivan, C. E. Newman et al. "Dust lifting through surface albedo changes at Jezero crater, Mars." *JGR: Planets* (2023) DOI: 10.1029/2022JE007672.
39. J. Pla-García, A. Munguira, S. Rafkin, C. Newman, et al. "Nocturnal turbulence at Jezero crater as determined from MEDA measurements and modeling." *JGR: Planets* (2023) DOI: 10.1029/2022JE007607.
40. M. de la Torre Juárez, A. Chavez, L. K. Tamppari, A. Munguira, G. Martínez et al. "Diurnal cycle of rapid air temperature fluctuations at Jezero crater: probability distributions, exponential tails, scaling, and intermittency." *JGR: Planets* (2023) DOI: 10.1029/2022JE007458.
41. S. Menina, L. Margerin, et al. "Stratification of Heterogeneity in the Lithosphere of Mars From Envelope Modeling of Event S1222a and Near Impacts: Interpretation and Implications for Very-High-Frequency Events." *GRL* (2023) DOI: 10.1029/2023GL103202.
42. B. Chide, X. Jacob, A. Petculescu, R. D. Lorenz et al. "Measurements of sound propagation in Mars' lower atmosphere." *Earth and Planetary Science Letters* (2023) DOI: 10.1016/j.epsl.2023.118200.
43. C. Alvarez-Llamas, J. Laserna, J. Moros, P. Purohit et al. "The sound of geological targets on Mars from the absolute intensity of laser-induced sparks shock waves." *Spectrochimica Acta Part B: Atomic Spectroscopy* (2023) DOI: 10.1016/j.sab.2023.106687.
44. L. Posiolova, P. Lognonné, W. B. Banerdt, J. Clinton, G. S. Collins, et al. "Largest recent impact craters on Mars: Orbital imaging and surface seismic co-investigation." *Science* (2022) DOI: 10.1126/science.abq7704.
45. N. Dahmen, J. F. Clinton, M-A. Meier, S. C. Stähler, S. Ceylan, D. Kim, **A. E. Stott**, and Domenico Giardini. "MarsQuakeNet: A more complete marsquake catalog obtained by deep learning techniques." *Journal of Geophysical Research: Planets* (2022) DOI: 10.1029/2022JE007503.
46. R. Hueso, C. E. Newman, T. del Río-Gaztelurrutia, A. Munguira, et al. "Convective vortices and dust devils detected and characterized by Mars 2020." *JGR: Planets* (2022) DOI: 10.1029/2022JE007516.
47. B. Chide, T. Bertrand, R. D. Lorenz, A. Munguira, et al. "Acoustics Reveals Short-Term Air Temperature Fluctuations Near Mars' Surface." *GRL* (2022) DOI: 10.1029/2022GL100333.
48. A. Horleston, J. F. Clinton, S. Ceylan, et al. "The Far Side of Mars: Two Distant Marsquakes Detected by InSight." *The Seismic Record* (2022) DOI: 10.1785/0320220007.
49. R. Wiens, A. Udry, O. Beyssac, et al. "Compositionally and density stratified igneous terrain in Jezero crater, Mars." *Science Advances* (2022) DOI: 10.1126/sciadv.abo3399.
50. S. Ceylan, J. F. Clinton, D. Giardini et al. "The marsquake catalogue from InSight, sols 0–1011." *Physics of the Earth and Planetary Interiors* (2022) DOI: 10.1016/j.pepi.2022.106943.
51. N. L. Dahmen, G. Zenhausern, J. F. Clinton, D. Giardini, S. C. Stahler, et al. "Resonances and lander modes observed by insight on Mars (1–9 Hz)." *BSSA* (2021) DOI: 10.1785/0120210056.
52. K. Hurst, L. Fayon, B. Knapmeyer-Endrun, C. Schmelzbach, M. van Driel, et al. "Resonances of the InSight seismometer on Mars." *BSSA* (2021) DOI: 10.1785/0120210137.
53. J. F. Clinton, S. Ceylan, M. van Driel, D. Giardini et al. "The Marsquake catalogue from InSight, sols 0–478." *Physics of the Earth and Planetary Interiors* (2021) DOI: 10.1016/j.pepi.2020.106595.

54. S. Ceylan, J. F. Clinton, D. Giardini, M. Bose et al. "Companion guide to the marsquake catalog from InSight, Sols 0–478: Data content and non-seismic events." *Physics of the Earth and Planetary Interiors* (2021) DOI: 10.1016/j.pepi.2020.106597.
55. C. Charalambous, J. McClean, M. Baker, W. T. Pike, et al. "Vortex-dominated aeolian activity at InSight's landing site, Part 1: Multi-instrument observations, analysis, and implications." *JGR: Planets* (2021) DOI: 10.1029/2020JE006757.
56. T. Lecocq, S. P. Hicks, K. Van Noten, K. Van Wijk et al. "Global quieting of high-frequency seismic noise due to COVID-19 pandemic lockdown measures." *Science* (2020) DOI: 10.1126/science.abd2438.

Selected Conferences

1. A. E. Stott, L. Margerin, et al. "Identification of new Marsquake clusters from P-wave coda analysis and implications for crustal structure and seismicity" Europlanet Science Conference (EPSC), (2026).
2. A. E. Stott, M. Marin, et al. "Life on the edge of a Martian storm track: extracting information on aeolian processes at Jezero from MEDA hardware failures on Mars 2020" Europlanet Science Conference (EPSC), (2026).
3. A. E. Stott, R. Garcia, et al. "Inferring Global Noise Maps for Future Seismology Missions to Mars" British Planetary Science Conference, (2026).
4. A. E. Stott, D. Mimoun, et al. "The Platform Atmospheric Characteristics Instrumentation Suite (PACIS) for sensing Mars's atmosphere during the entry, descent and landing of the ESA ExoMars Rosalind Franklin mission" Europlanet Science Conference (EPSC), (2025).
5. A. E. Stott, R. F. Garcia, M. Drilleau, L. Margerin et al. "Examining the P-wave coda features of InSight seismic events" EGU General Assembly (2025)
6. A. E. Stott, N. Murdoch, M. Gillier et al. "The Distribution of Short-term Wind Gusts and the Dissipation Regime of Turbulence on Mars with the SuperCam Microphone." Europlanet Science Conference (EPSC), (2024).
7. A. E. Stott, R. F. Garcia, N. Murdoch, D. Mimoun et al. "A Catalogue of Martian Winds Inferred from InSight's Seismic Data Using Machine Learning." Tenth International Conference on Mars (2024).
8. A. E. Stott, R. F. Garcia, N. Murdoch, D. Mimoun et al. "Noise Considerations For the Next InSight Into the Interior of Mars ." Mars Interior and Geophysics after InSight (2024).
9. A. E. Stott, G. Martinez, C. E. Newman, M. J. Wolff, M. de la Torre Juarez, and J. A. Rodriguez-Manfredi. "Atmospheric science with the M2020 Perseverance rover: key results and future objectives." British Planetary Science Conference (2024).
10. A. E. Stott and R. F. Garcia. "Analysing marsquakes and atmospheric noise with machine learning." Cargese passive imaging summer school (2024).
11. A. E. Stott, R. F. Garcia, N. Murdoch, D. Mimoun et al. "Using InSight data to inform sensing opportunities for future seismology and meteorology missions" International Planetary Probe Workshop (2023).
12. A. E. Stott, N. Murdoch, M. Gillier et al. "Martian Wind and turbulence heard by the SuperCam microphone on the Perseverance rover" Acoustical society of America, Invited talk (2023).
13. A. E. Stott, R. F. Garcia, A. Chédozeau, B. Pinot, N. Murdoch, D. Mimoun, et al. "Using machine learning to separate atmospherically generated noise from marsquakes" EPSC (2022) EPSC2022-833.
14. A. E. Stott, N. Murdoch, R. F. Garcia, M. Gillier, D. Mimoun, et al. "Martian Wind Sensing with Seismometers and Microphones." Seventh conference on Mars Atmospheric Modelling and observations (2022).
15. A. E. Stott, R. F. Garcia, A. Chédozeau, B. Pinot, N. Murdoch, D. Mimoun, et al. "Using machine learning to predict the ground motion from atmospheric data at InSight, Mars." AGU (2021).
16. A. E. Stott, N. Murdoch, D. Mimoun, B. Chide, R. Lorenz, S. Maurice, et al. "The Sound of the Wind on Mars: Preliminary Wind Speed Analysis with the SuperCam Microphone on Perseverance." EPSC (2021): EPSC2021-557.
17. A. E. Stott, R. F. Garcia, B. Pinot, N. Murdoch, D. Mimoun, A. Spiga, et al. "Atmospherically driven ground motion at InSight: a machine learning perspective." EGU General Assembly (2021) EGU21-12344.
18. A. E. Stott, C. Charalambous, J. B. McClean, T. Warren, A. Trebi-Ollennu, G. Lim, N. Teanby et al. "Using InSight's Robotic Arm Motion to Examine the Martian Regolith's Response to Short Period Vibrations." LPI 2326 (2020): 2082.
19. A. E. Stott, C. Charalambous, W. T. Pike, J. McClean, T. Warren, R. F. Garcia, P. Lognonné et al. "Comodulation Analysis of Atmospheric Energy Injection into the Ground Motion at InSight Part 1: The Site Dynamics on Elysium Planitia." AGUFM 2019 (2019): DI51A-0004.
20. W. T. Pike, P. Lognonné, W. B. Banerdt, S. B. Calcutt, I. M. Standley, D. Giardini, C. Charalambous et al. "Results from the Short-Period (SP) Seismometers on the Mars Insight Mission: From Launch to Sol 40." In Lunar and Planetary Science Conference, vol. 50. 2019.
21. W. T. Pike, S. Calcutt, I. M. Standley, A. G. Mukherjee, J. Temple, T. Warren, C. Charalambous, H. Liu, A. E. Stott, and J. B. McClean. "A silicon seismic package (SSP) for planetary geophysics." In: Lunar and Planetary Science Conference, vol. 47, 2016, p. 2081.